

Project Proposal

INF2167 - R for Data Science

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Introduction

Toronto aims to expand its tree canopy cover from its current 28-31% to 40% by 2050 to build climate resilience and mitigate urban heat (City of Toronto 2025). To evaluate if current planting strategies will actually secure this future canopy, this project analyzes municipal street tree data (City of Toronto 2026) against three foundational indicators of urban forest stability: taxonomic diversity, structural age distribution, and mortality risk. A resilient canopy must avoid monoculture vulnerability by adhering to the 10-20-30 rule, ensuring no single species, genus, or family exceeds 10%, 20%, or 30% of the population (Freiburger et al. 2025). Structurally, sustainable population replacement requires a “youthful” (Type I) distribution, characterized by a “reverse-J” curve where abundant young trees (measured by DBH) are recruited to replace mature losses (Morgenroth et al. 2020). However, because young trees face severe mortality risks within their first five years (Hilbert et al. 2019), this youthful distribution only translates into a mature canopy if paired with rigorous early stewardship. Identifying these structural and demographic bottlenecks is critical to ensure Toronto’s forestry investments yield a sustainable, long-term canopy rather than a costly cycle of early mortality.

Research Questions

By evaluating the combined impacts of taxonomic diversity, size distribution, and early mortality, this project investigates whether Toronto’s street tree planting efforts can realistically support the city’s goal of 40% canopy cover by 2050. It is hypothesized that despite high planting volumes, a demographic bottleneck caused by harsh roadside conditions leads to disproportionate early mortality among young trees. Consequently, because these trees die before growing large enough to provide meaningful shade, the current street tree population is likely failing to generate the mature growth necessary to achieve the 2050 target.

Dataset description, limitations, and methods

The primary dataset, “Street Tree Data” from Toronto’s Open Data portal, provides an inventory of predominantly City-owned trees on road allowances. Compiled during Urban Forestry maintenance, the dataset contains over 530,000 records with key variables including DBH_TRUNK, COMMON_NAME, BOTANICAL_NAME, WARD, and geometry. Limitations to account for include the exclusion of private and park trees (which may underestimate overall canopy health), outdated records tied only to active maintenance, and imprecise parcel-level geocoding instead of exact GPS coordinates.

To evaluate if this population can support Toronto’s 2050 canopy target, R will be used to model taxonomic and demographic structure (R Core Team 2024), along with the R packages, tidyverse (Wickham et al. 2019) and here (Müller 2020) . Diversity will be assessed against the 10-20-30 rule using BOTANICAL_NAME. Using DBH_TRUNK as a proxy for age, size distributions for the five most common species will then be analyzed to confirm sustainable “reverse-J” replacement curves and to identify early mortality bottlenecks in low-DBH cohorts.

Exploratory Data Analysis

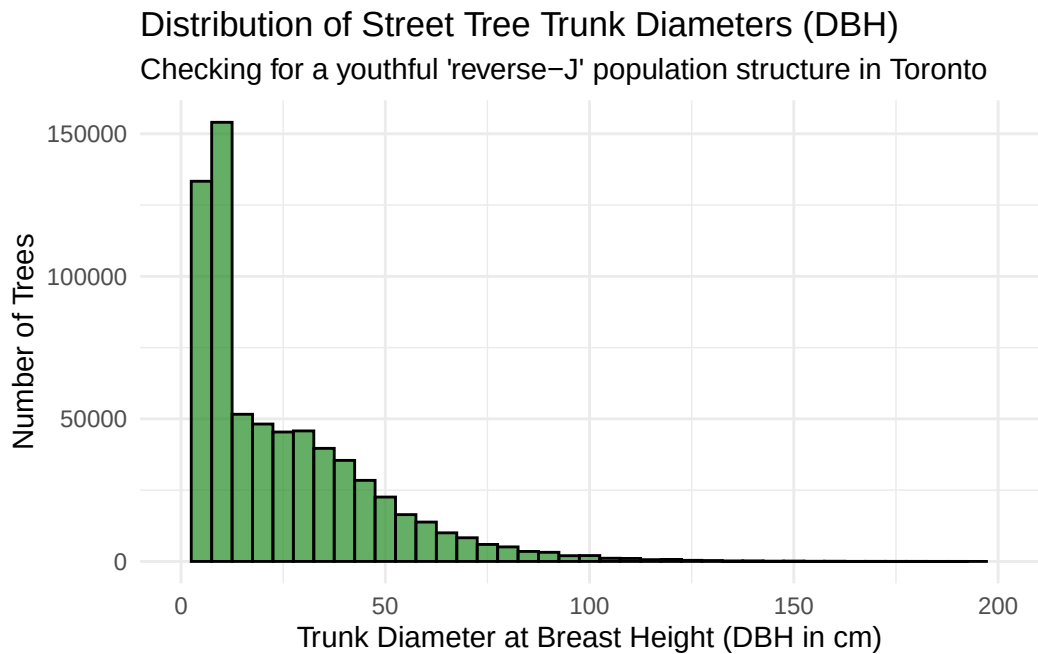
```
# Load tidyverse library
library(tidyverse)
# here library used for reproducible file paths
library(here)

# Load the locally saved CSV data
tree_data <- read_csv("Street_Tree_Data.csv")

# Removing rows that contain impossible value of DBH
clean_trees <- tree_data %>%
  filter(DBH_TRUNK > 0) %>%
  drop_na(COMMON_NAME, DBH_TRUNK)

# Creating a histogram to visualize if the DBH distribution contains a reverse J-curve
ggplot(clean_trees, aes(x = DBH_TRUNK)) +
  geom_histogram(binwidth = 5, fill = "forestgreen", color = "black", alpha = 0.7) +
  theme_minimal() +
  xlim(0, 200) +
  labs(
    title = "Distribution of Street Tree Trunk Diameters (DBH)",
    subtitle = "Checking for a youthful 'reverse-J' population structure in Toronto",
    x = "Trunk Diameter at Breast Height (DBH in cm)",
```

```
y = "Number of Trees"
)
```

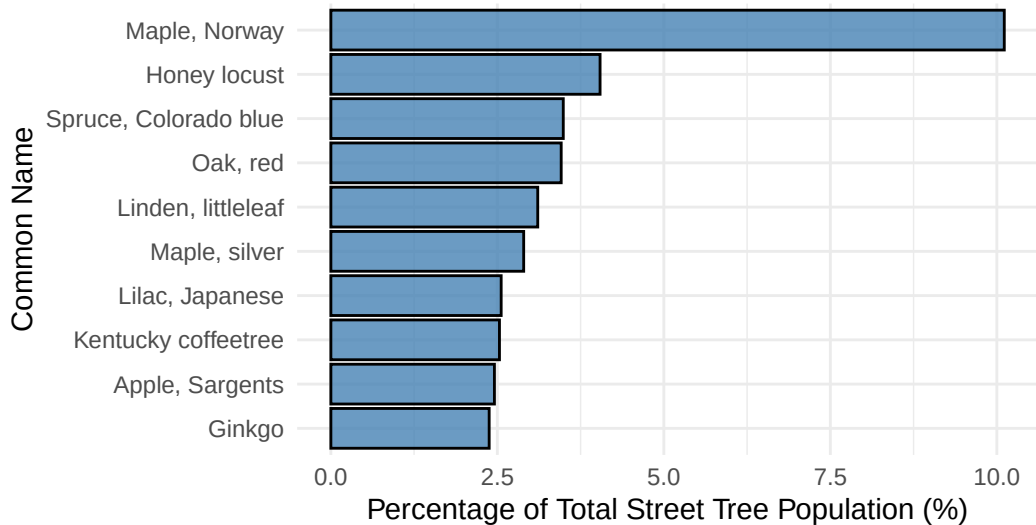


```
# This graph plots the top 10 most common species to determine monoculture vulnerability
species_dominance <- clean_trees %>%
  count(COMMON_NAME) %>%
  mutate(Percentage = (n / sum(n)) * 100) %>%
  slice_max(Percentage, n = 10)

# This is plotted in a bar chart
ggplot(species_dominance, aes(x = reorder(COMMON_NAME, Percentage), y = Percentage)) +
  geom_col(fill = "steelblue", color = "black", alpha = 0.8) +
  coord_flip() +
  theme_minimal() +
  labs(
    title = "Top 10 Most Dominant Species in Toronto",
    subtitle = "Evaluating over-reliance on specific species and \nmonoculture vulnerability",
    x = "Common Name",
    y = "Percentage of Total Street Tree Population (%)"
  )
```

Top 10 Most Dominant Species in Toronto

Evaluating over-reliance on specific species and monoculture vulnerability



The exploratory data analysis investigated the structural and taxonomic composition of Toronto's street trees to assess the viability of the 2050 canopy target. Analyzing the diameter at breast height (DBH) reveals a highly skewed, youthful population distribution; while the pronounced "reverse-J" curve demonstrates aggressive recent planting efforts, the precipitous drop-off in tree frequency beyond the 15 cm DBH threshold exposes a severe early-mortality bottleneck. This stark decline suggests that despite high planting volumes, young trees are largely failing to survive harsh roadside conditions long enough to mature and provide meaningful shade. Concurrently, the taxonomic evaluation indicates that the city's street tree population narrowly adheres to the 10-20-30 diversity rule at the species level. The Norway Maple constitutes precisely 10% of the population, sitting precariously at the maximum allowable threshold for a single species, which flags a localized vulnerability to monoculture risks and species-specific pests.

References

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